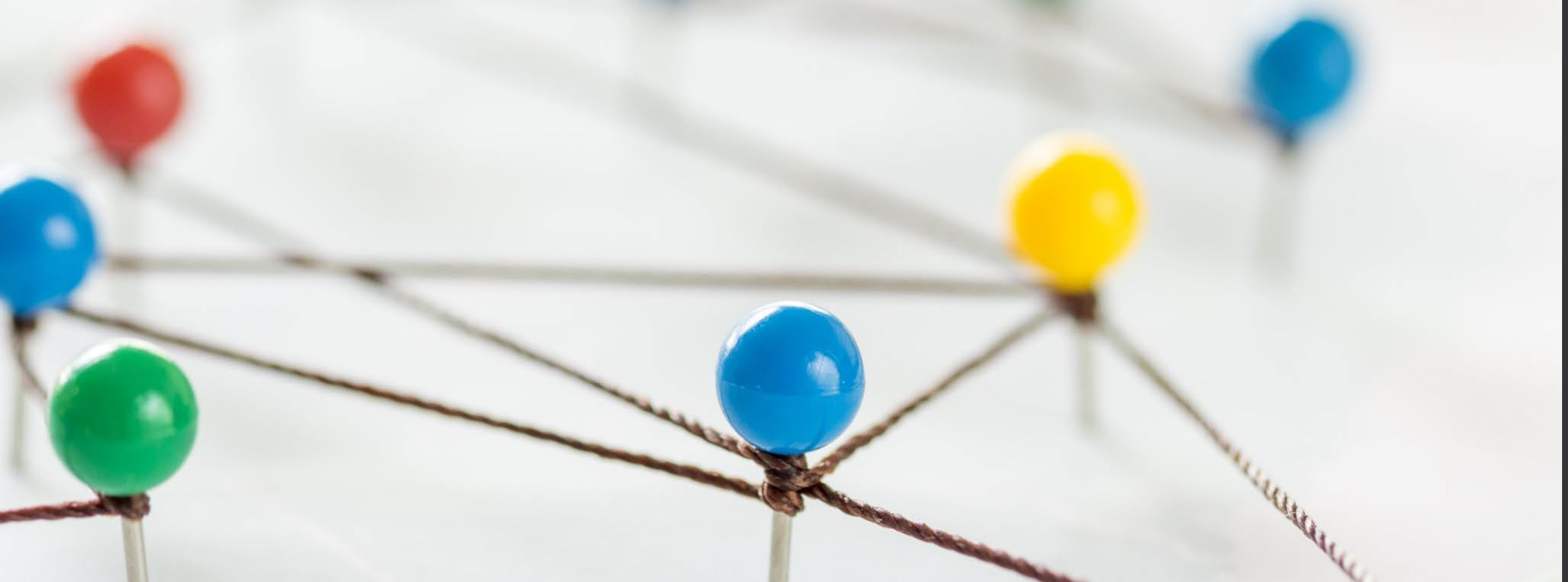


S. Mason Garrison

PROBABILITY

*May the odds be ever in
your favor*



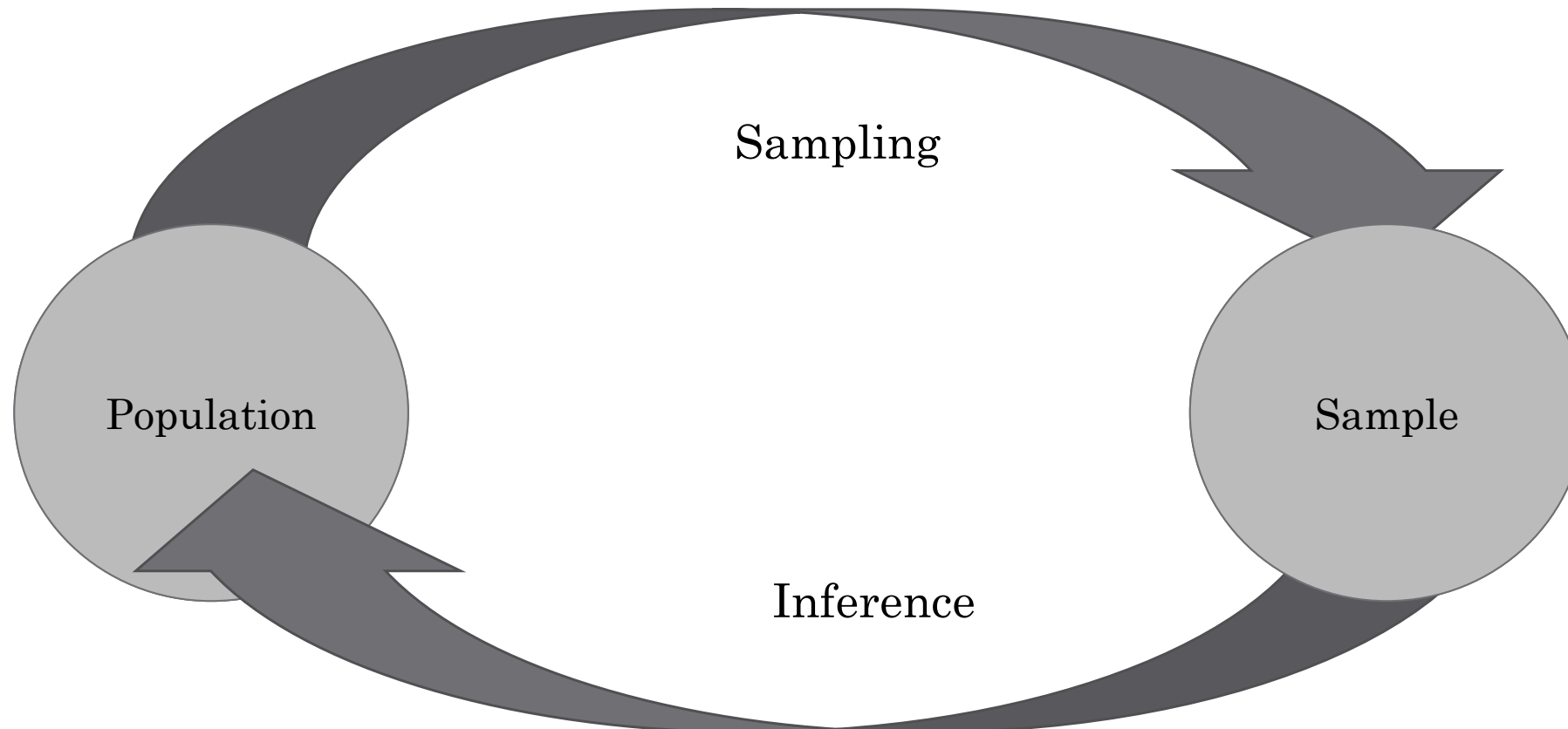


Road Map

- Goals for Inference
- History of Probability
- Definitions
- Probability Models

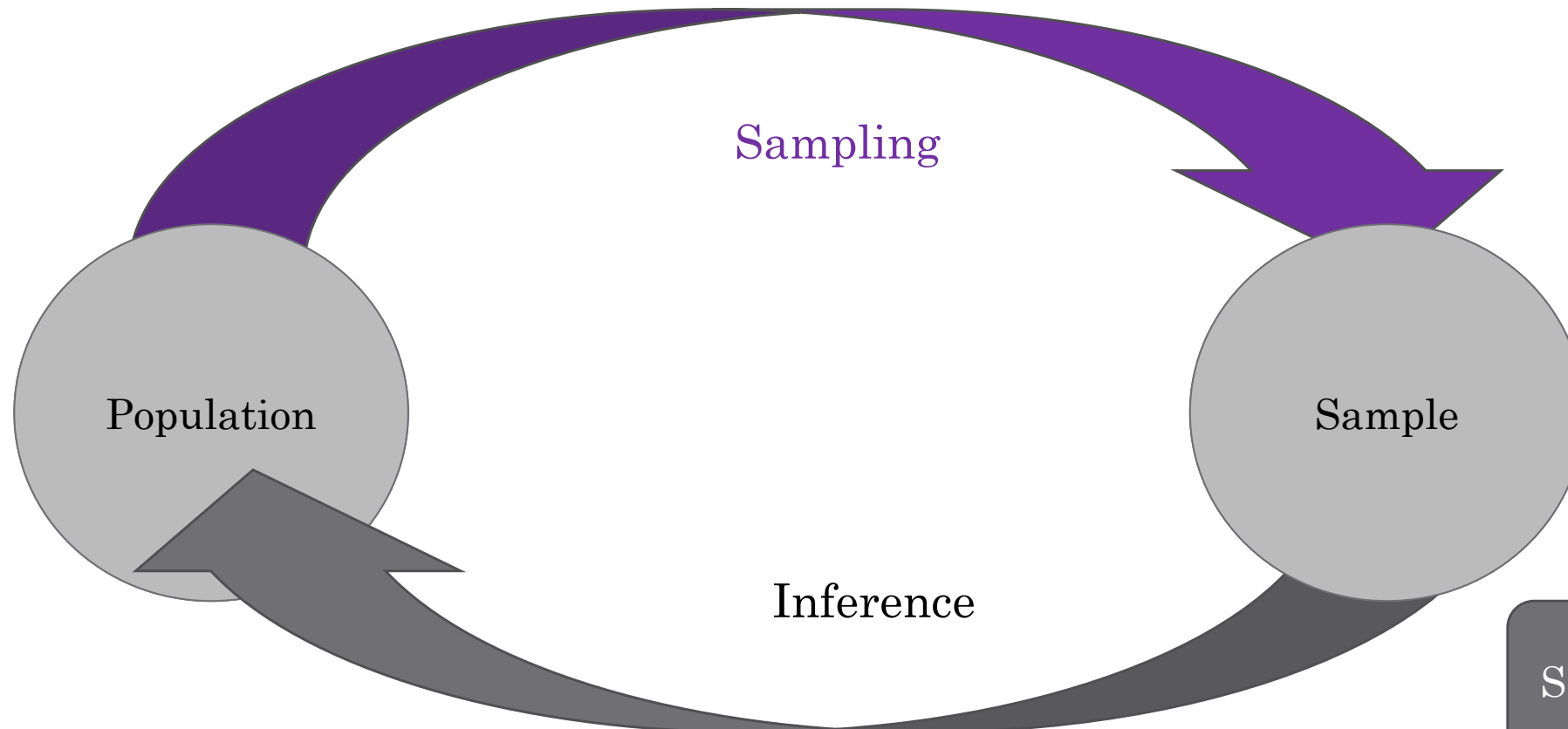
The bridge model of statistical inference – confirmatory data analysis (CDA)

- In EDA, we build theories, based on evidence, of how the world works.
- In CDA, we test theories based on procedures (statistical tests, etc)



The bridge model of statistical inference – confirmatory data analysis (CDA)

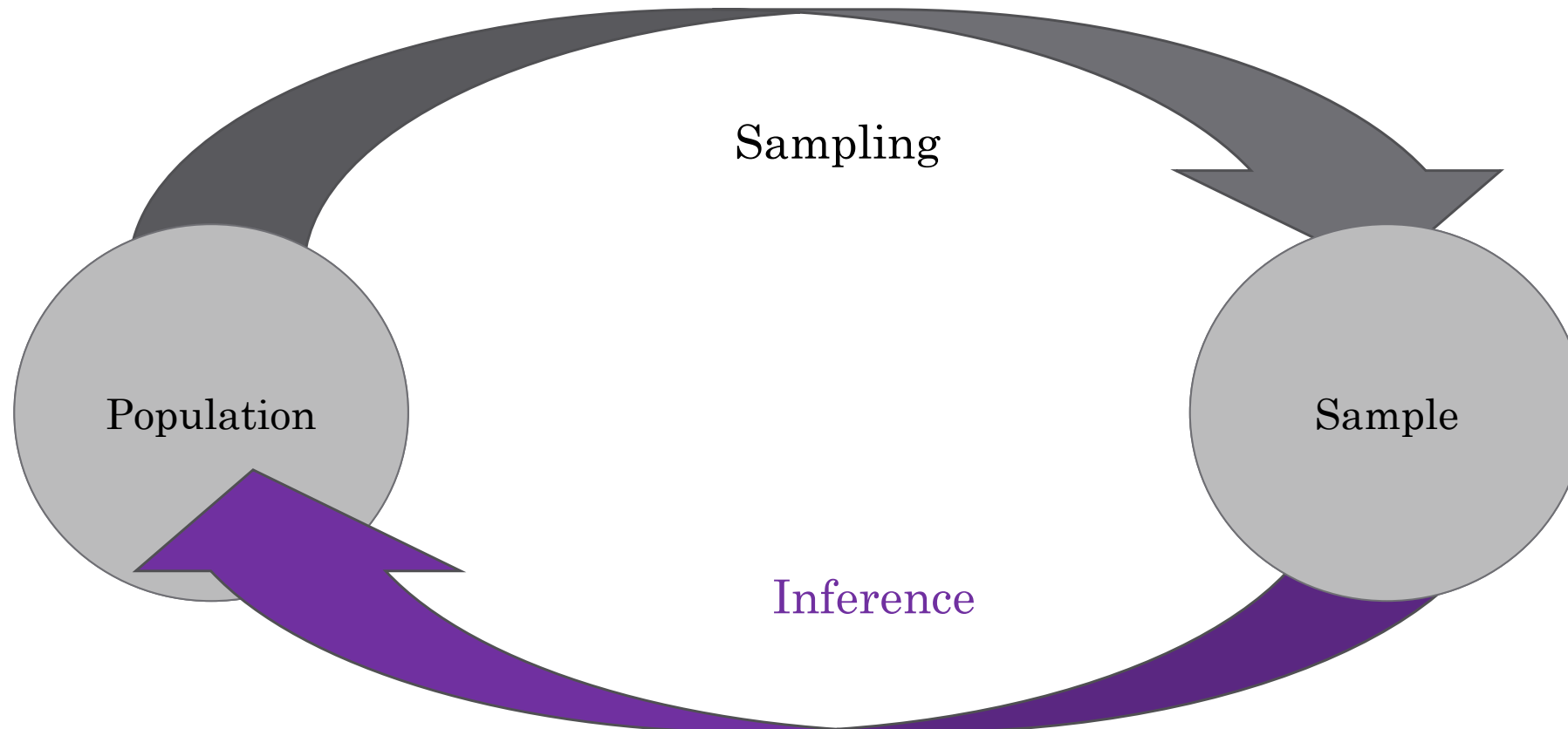
- The population can be infinite, while the sample cannot be
- Sampling theory allows us to cross the bridge from population to sample



Note: Convenience Sampling doesn't build that bridge

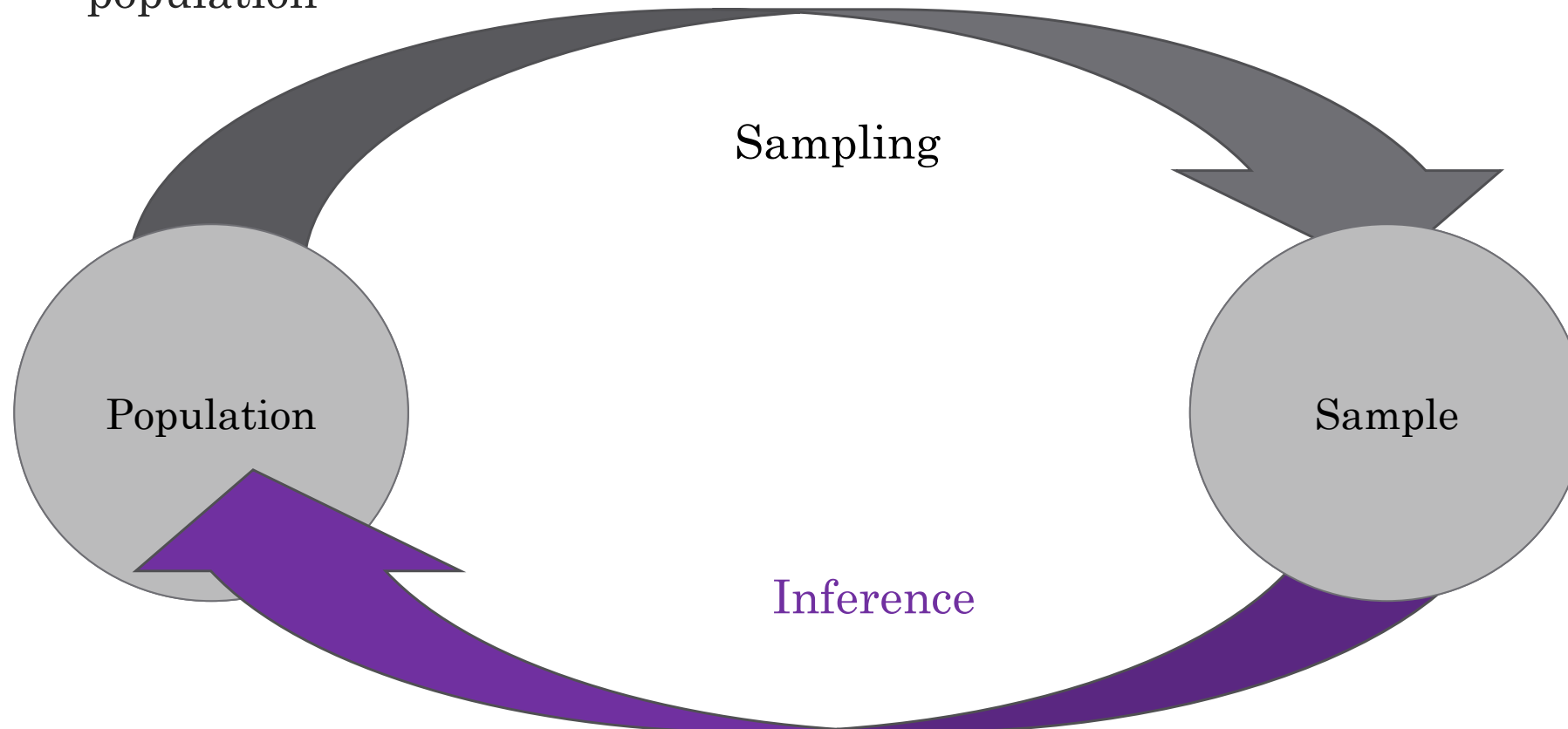
The bridge model of statistical inference – confirmatory data analysis (CDA)

- Probability theory allows us to cross the bridge from the sample to the population



The bridge model of statistical inference – confirmatory data analysis (CDA)

- Only works if sampling method was good.
- Researcher can only see the sample, use the sample to understand the population



Goal

make statements about the
population from the sample



estimating population
parameters



test hypotheses





Wrapping Up...

This Time...



History of Probability

- **France, 1654: Antoine Gombaud, Chevalier de Méré**
 - French nobleman, gambler, and socialite, who was
 - famous for hosting gambling parties and games of chance

Louis XIV, the Sun King, has his coronation this same year, after 11 years on the Throne



History of Probability

- **France, 1654: Antoine Gombaud, Chevalier de Méré**
 - French nobleman, gambler, and socialite, who was
 - famous for hosting gambling parties and games of chance
- **The Contradiction in Dice Games**
- Chevalier de Méré noticed a puzzling contradiction in two popular dice games, which led to one of the first mathematical inquiries into probability:
 - Game 1: Bet on rolling a 6 in four rolls.
 - Game 2: Bet on rolling two 6s in twenty-four rolls.





What He Believed

- Based on a well-established gambling rule,
 - de Méré believed that betting on two sixes in 24 rolls would be profitable.
- However, despite his calculations, this strategy turned out not to be profitable.
 - Extremely, NOT profitable

Turning to Experts: Pascal and Fermat



He asked Pascal and Fermat to explain the contradiction



These letters between
Pascal and Fermat are

probably the first formal
theory of probability



Shortly after, Dutch scientist **Christian Huygens** learned of the letters between Pascal and Fermat.



In 1657, he published the first book on probability, titled *De Ratiociniis in Ludo Aleae* (On Reasoning in Games of Chance).



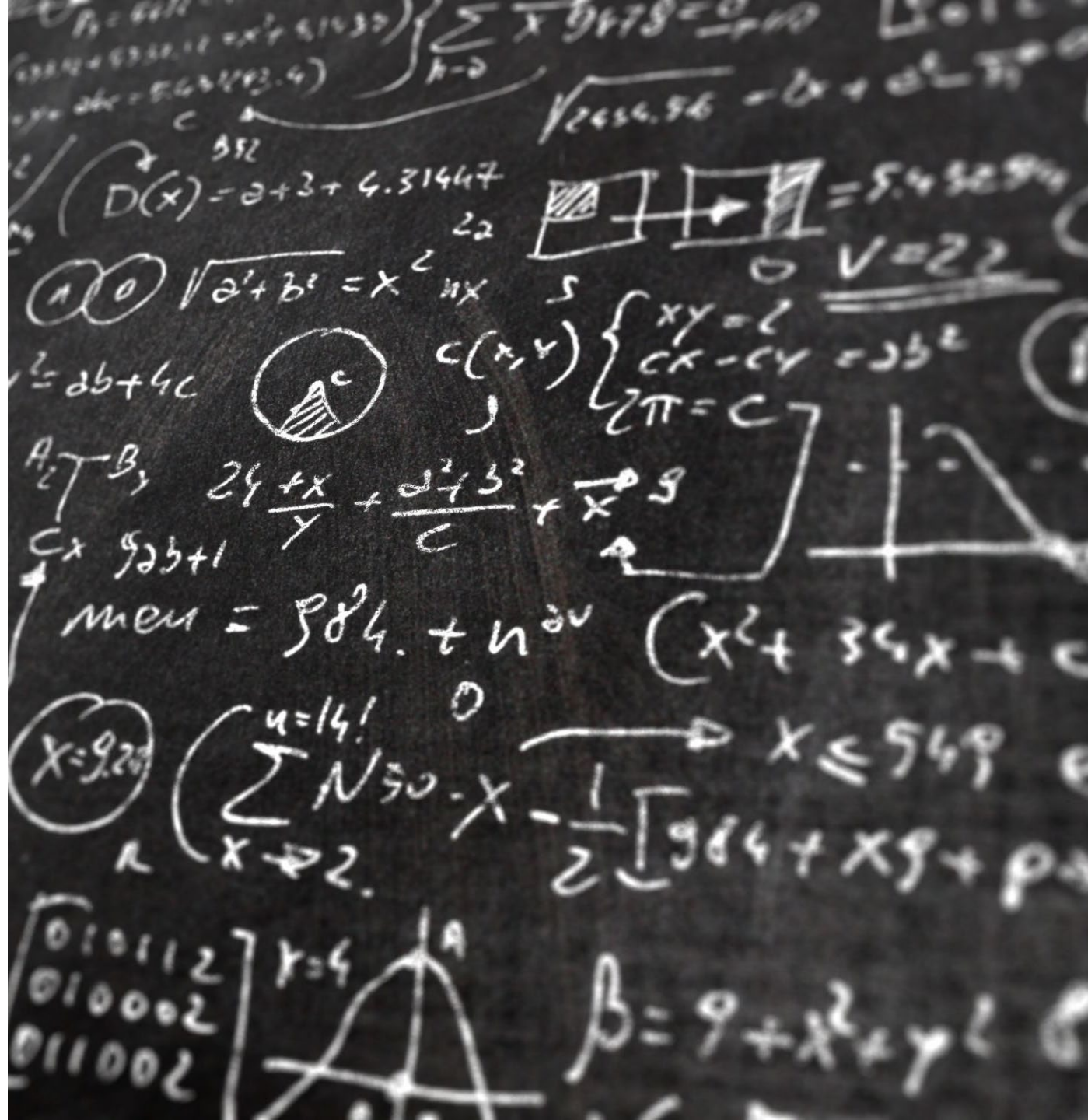
Fun Fact: Christian Huygens was a teacher of Leibniz, one of the most prominent philosophers and mathematicians in history.

Christian Huygens (1657)



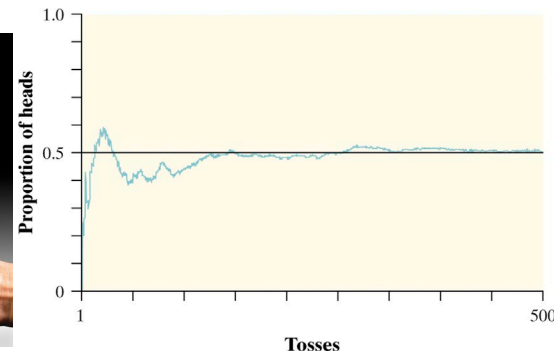
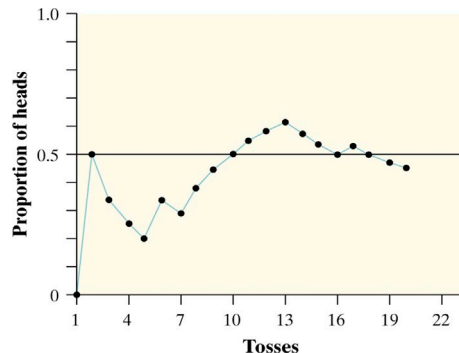
Wrapping Up...

Probability Definitions



Probability Definitions

- Big Idea: Chance behavior is unpredictable in the short run, but it has a regular and predictable pattern in the long run.
- Simple def. Way to quantify uncertainty
- Textbook def. Mathematics of chance behavior
- Operational def. The probability of any outcome of a random phenomenon is the proportion of times the outcome would occur in a very long series of repetitions.
 - A phenomenon is random if individual outcomes are uncertain but there is, nonetheless, a regular distribution of outcomes in a large number of repetitions



More Definitions

- Event – outcome or set of outcomes of a random phenomenon
 - an event is a subset of the sample space
- Sample space – set of all possible outcomes
 - Example : coin flip
 - Possible outcomes: heads or tails
 - $\{H, T\}$
 - Die roll
 - $\{1, 2, 3, 4, 5, 6\}$
 - Cards in a deck
 - {Ace of clubs... 2 of clubs,..., A hearts, A spades, etc}



Even More Definitions

Probability model

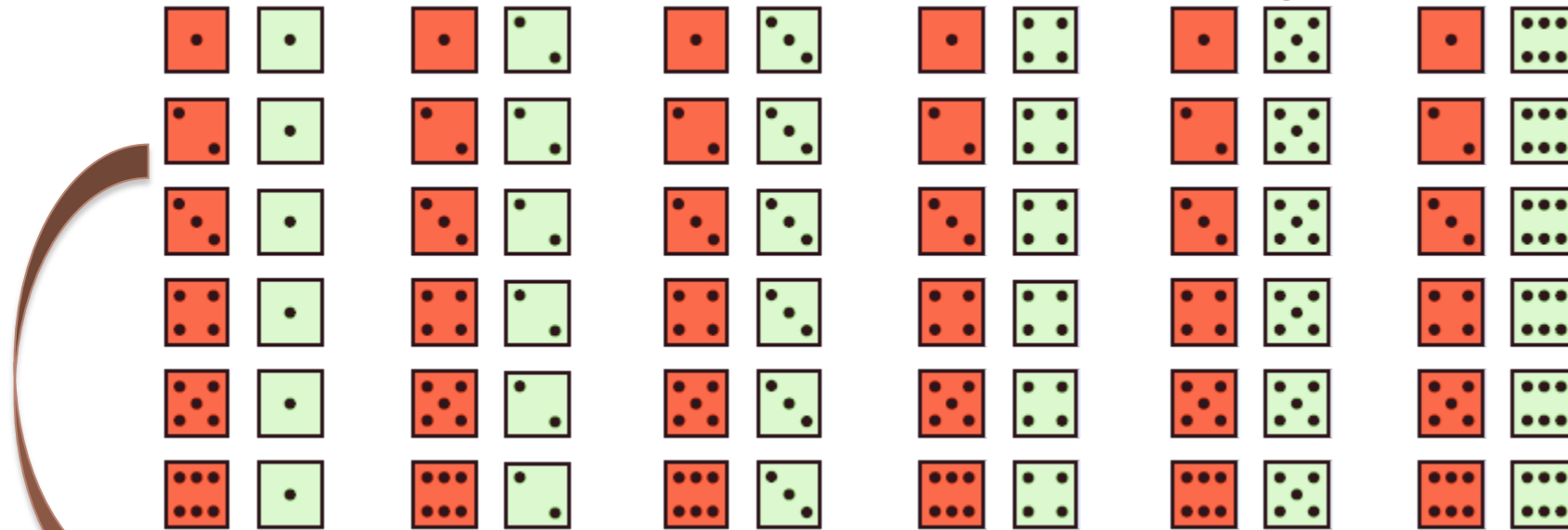
- a mathematical description of a random phenomenon consisting of two parts:
 - a sample space S and
 - a way of assigning probabilities to events

Classical def of probability

- A is a set of outcomes in sample space S
- $P(A) = \# \text{ of events in } A / \# \text{ of events in } S$
- Example
 - $P(\text{king}) = 4/52 = 1/13$
 - $P(\text{king of spades}) = 1/52$

Probability Example

Example: Give a probability model for the chance process of rolling two fair, six-sided dice—one that is red and one that is green.



**Sample space
36 outcomes**

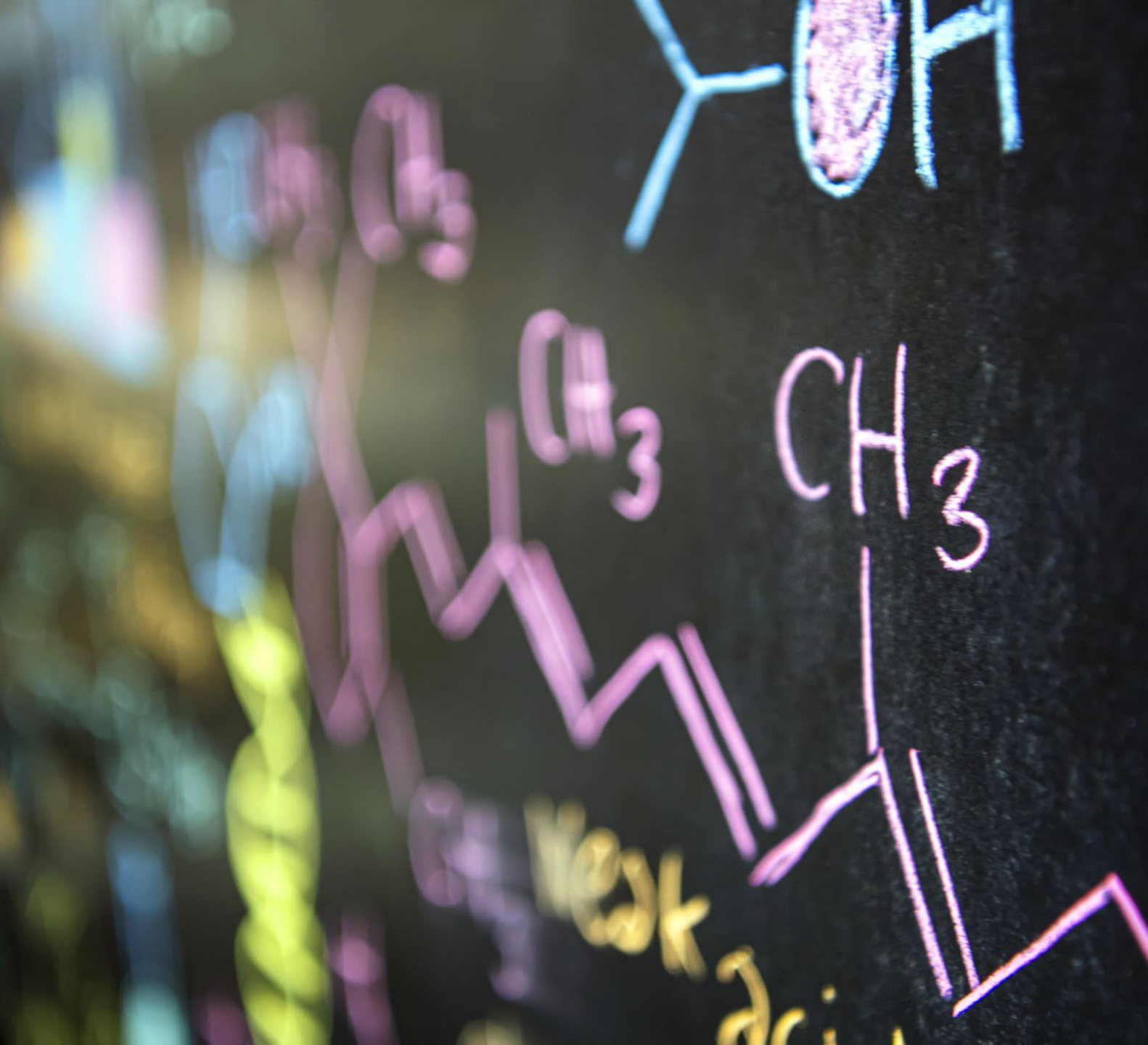
Since the dice are fair, each outcome is equally likely.
Each outcome has probability $1/36$.

Properties of Probabilities

- Any probability is a number between 0 and 1.
 - $0 \leq P(A) \leq 1$

Properties of Probabilities

- All possible outcomes together must have probability 1.
 - Because some outcome must occur on every trial,
 - the sum of the probabilities for all possible outcomes must be exactly 1
 - $P(S) = 1$
 - Probability of sample space must be 1

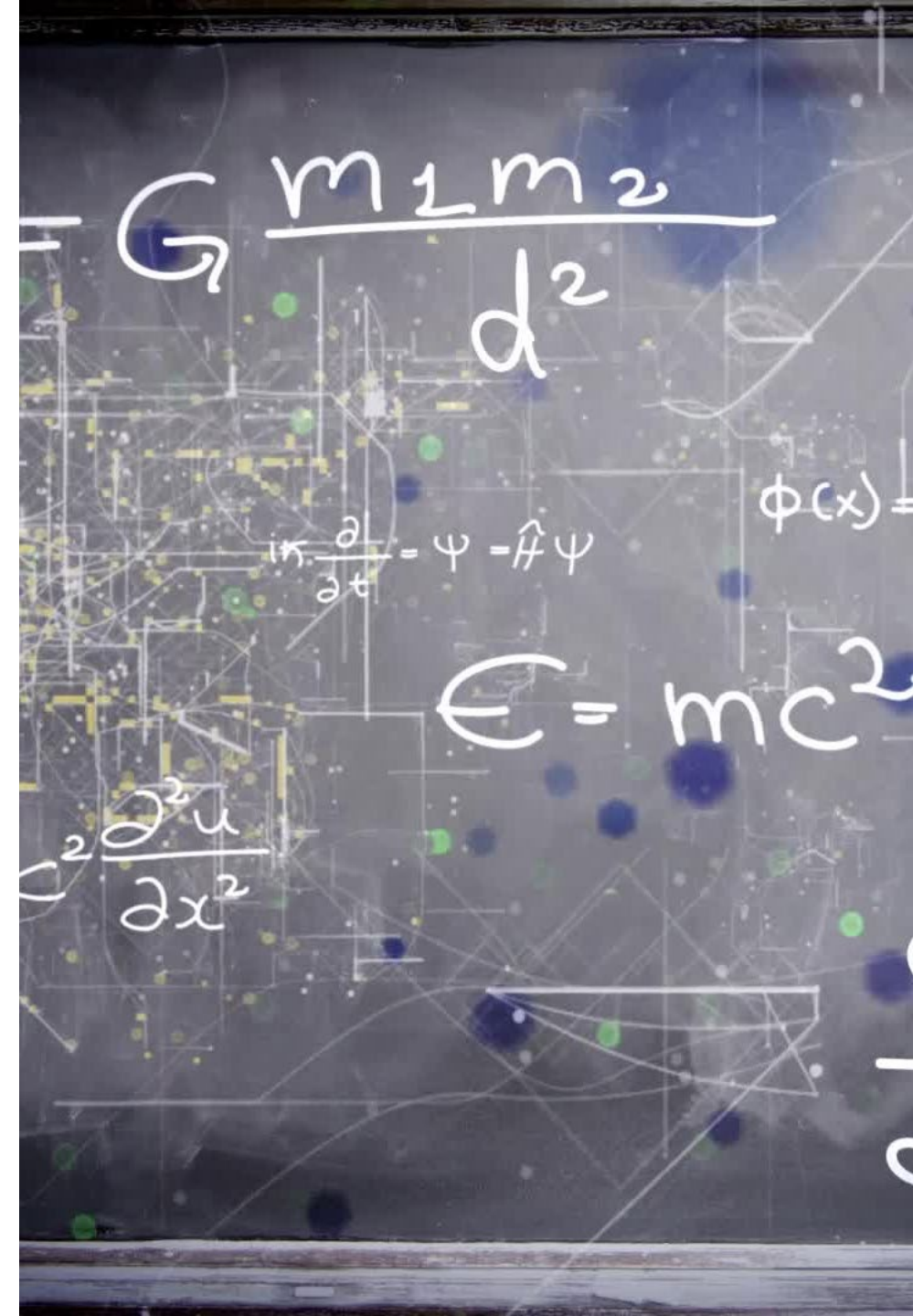
A chalkboard with various chemical structures and formulas drawn in different colors of chalk. At the top, there is a blue Y-shaped structure and a pink oval structure. Below them, there are several pink structures, including a zigzag line and a structure labeled 'CH3'. At the bottom, there is a yellow structure and a pink structure. The background is dark and textured.

Properties of Probabilities

- If two events have no outcomes in common (disjoint), the probability that one or the other occurs is the sum of their individual probabilities.
 - $P(A \text{ or } B) = P(A) + P(B)$
IF disjoint
 - Disjoint (A and B don't overlap)

Properties of Probabilities

- The probability that an event does not occur is 1 minus the probability that the event does occur.
 - The probability that an event occurs and the probability that it does not occur always add to 1, or 100%.
 - $P(\text{not } A) = 1 - P(A)$



Recap: Abridged Properties of Probabilities

- $0 \leq P(A) \leq 1$
- $P(S) = 1$
 - Probability of sample space must be 1
- $P(A \text{ or } B) = P(A) + P(B)$ IF disjoint
 - Disjoint (A and B don't overlap)
- $P(\text{not } A) = 1 - P(A)$



Probability in Action (example)

- What is Benford's Law?
 - In many naturally occurring datasets, the first digit of numbers follows a predictable distribution.
 - Instead of each digit being equally likely, smaller digits, especially 1, occur much more frequently as the first digit.
- Where do we see this?
 - Legitimate financial records, populations, stock prices, and even psychological data.



Probability rules (example)

- The first digits of numbers in legitimate financial records often follow a model known as Benford's law.
- Call the first digit of a randomly chosen record X .
- Benford's law gives the following probability model for X .

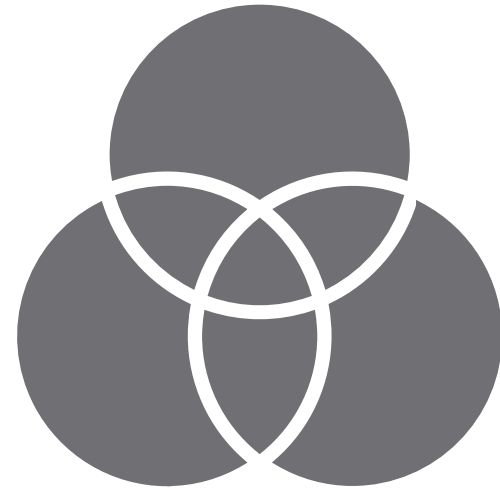
First digit (X)	1	2	3	4	5	6	7	8	9
Probability	0.301	0.176	0.125	0.097	0.079	0.067	0.058	0.051	0.046

How does this relate to the probability rules?

Benford's Law is a real-world application of the probability properties we've just reviewed.

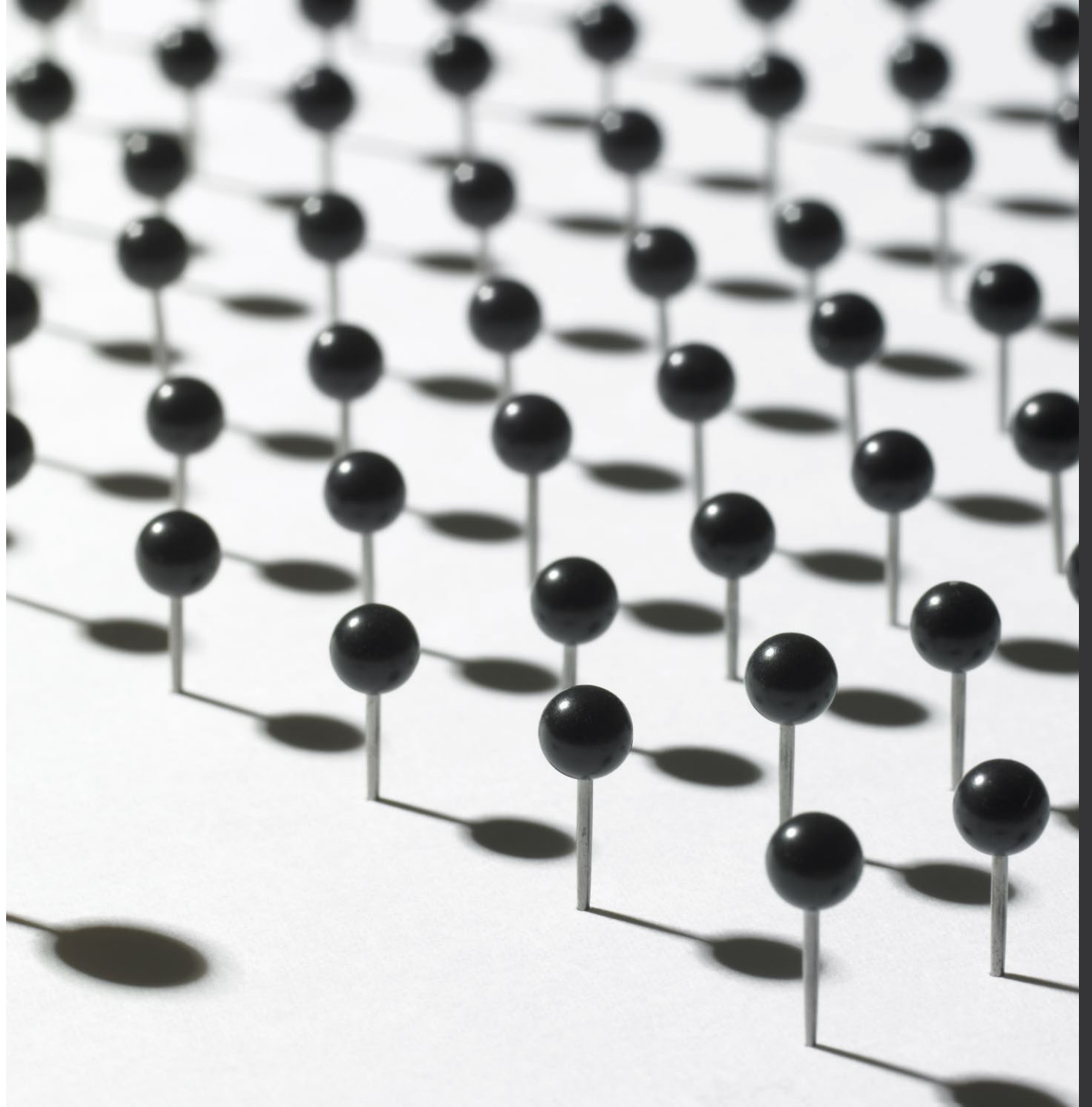
Your Task: How can we verify that this is a legitimate probability model?

Or is it one that I just made up?... Like the fake financial records that this purports to detect?



Wrapping Up...

Probability
Models...



A decorative background on the left side of the slide featuring a grid of small, colorful stars (red, yellow, green, blue, and silver) scattered across a light-colored surface.

Finite probability models

- One way to assign probabilities to events is to assign a probability to every individual outcome,
 - then add these probabilities to find the probability of any event.
 - This idea works well when there are only a finite (fixed and limited) number of outcomes.
- A probability model with a finite sample space is called a finite probability model.

Finite probability models

To assign probabilities in a finite model,

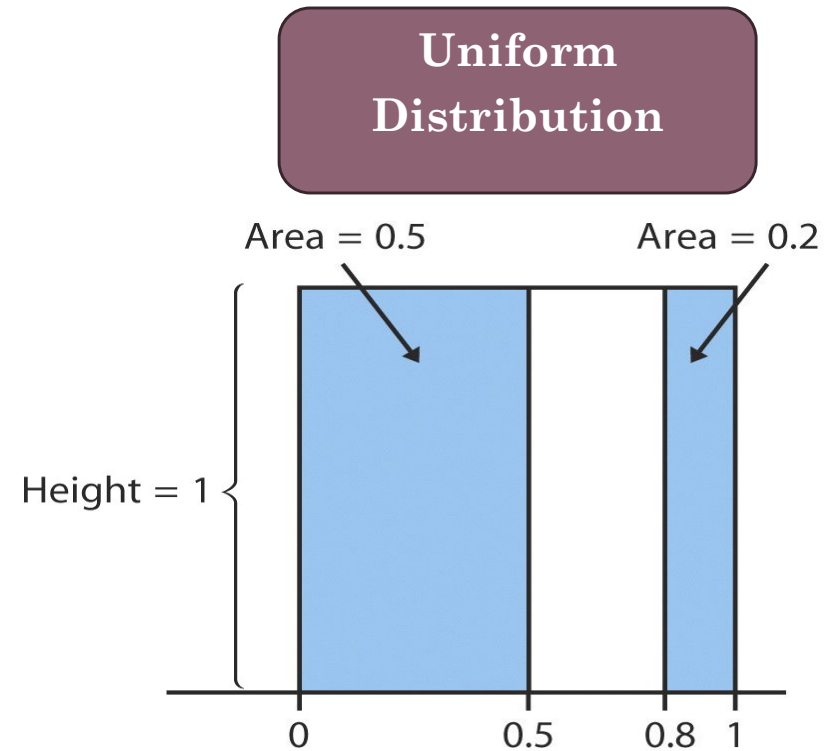
- list the probabilities of all the individual outcomes.
- These probabilities must be numbers between 0 and 1 that add to exactly 1.
- The probability of any event is the sum of the probabilities of the outcomes making up the event.

Finite probability models are sometimes called discrete probability models. Statisticians often refer to finite probability models as discrete.

Benford's law probability model was finite.

Continuous probability models

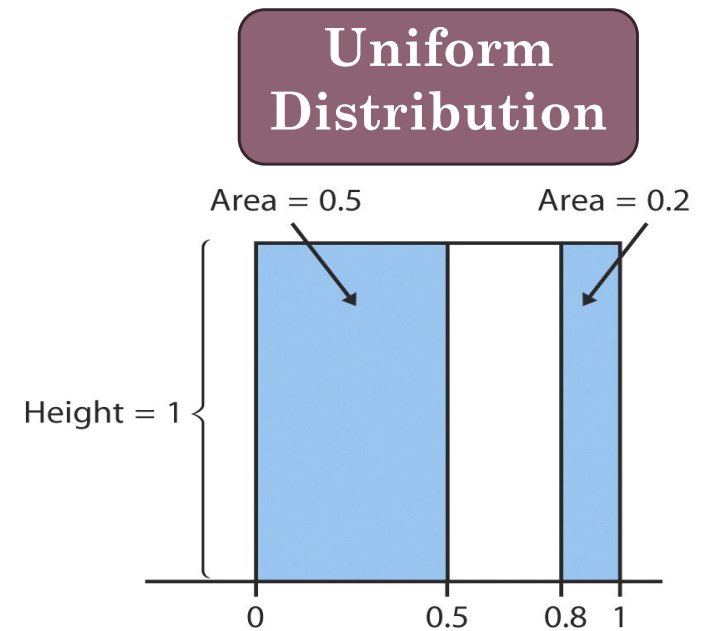
- Suppose we want to choose a number at random between 0 and 1, allowing any number between 0 and 1 as the outcome.
- We cannot assign probabilities to each individual value because there is an infinite continuum of possible values.
- A continuous probability model assigns probability as an area under a density curve. The area under the curve and above any range of values is the probability of an outcome in that range.



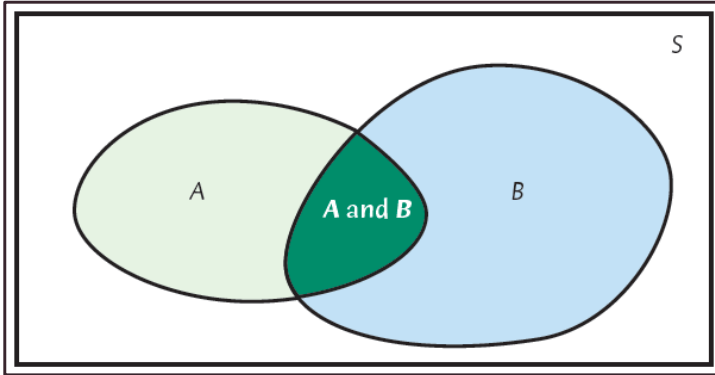
Continuous probability models

Example: Find the probability of getting a random number that is less than or equal to 0.5 or greater than 0.8.

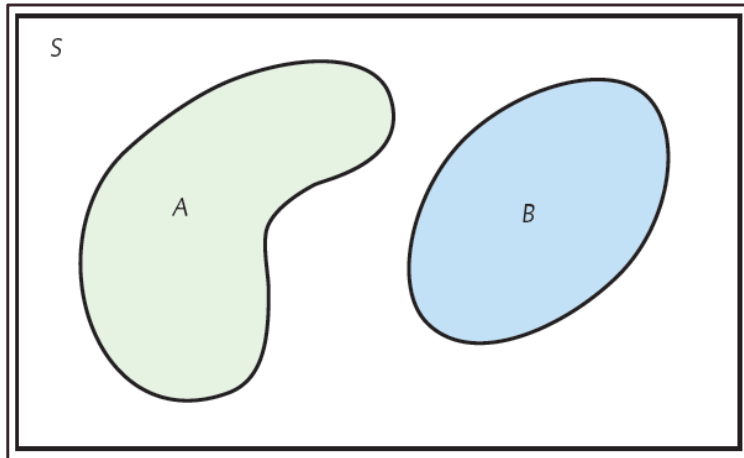
$$\begin{aligned} &P(X \leq 0.5 \text{ or } X > 0.8) \\ &= P(X \leq 0.5) + P(X > 0.8) \\ &= 0.5 + 0.2 = 0.7 \end{aligned}$$



Two events that are not disjoint.
The event $\{A \text{ and } B\}$ consists of the outcomes that they have in common.



Two events that are disjoint.



Multiple Events

- The upcoming rules look at multiple events.
 - I find it helpful to draw a picture to display relations among several events.
- A picture that shows the sample space S as a rectangular area and shows events as areas within S is called a Venn diagram.

Statistical independence

- Two events are independent if $P(A | B) = P(A)$, $P(B | A) = P(B)$
- Two events A and B are independent if knowing that one occurs does not change the probability that the other occurs
- Example
 - $A = \text{heart}$; $B = \text{jack}$
 - $P(H) = 13/52$
 - $P(H | J) = 1/4$; are independent
- Be careful not to confuse disjointness and independence.
 - If A and B are disjoint, then the fact that A occurs tell us that B cannot occur—very dependent!



The probability we assign to an event can change if we know that some other event has occurred.

- If the occurrence of event B, depends in some sense on the occurrence of event A,
 - then we can talk about the conditional probability of B, given A.

$P(B | A)$

- B given A

Conditional Probabilities



Conditional Probabilities Examples

- Weather Scenario
 - B = Event: John carries an umbrella.
 - A = Condition: The weather forecast predicts rain.
- Key Idea: $P(B|A)$ = Probability that John carries an umbrella given the forecast predicts rain.
 - Important: $P(B|A) \neq P(A)P(B|A) \neq P(A)$. They are not the same, but the forecast may make it more likely that John carries an umbrella.

Conditional Probabilities Examples

- Drawing a King
 - $P(K) = 4/52$
 - $P(\text{2}^{\text{nd}} \text{ card is a king} \mid \text{1}^{\text{st}} \text{ card was a king}) = 3/51$
 - $P(\text{2}^{\text{nd}} \text{ card is a king} \mid \text{1}^{\text{st}} \text{ card not king}) = 4/51$

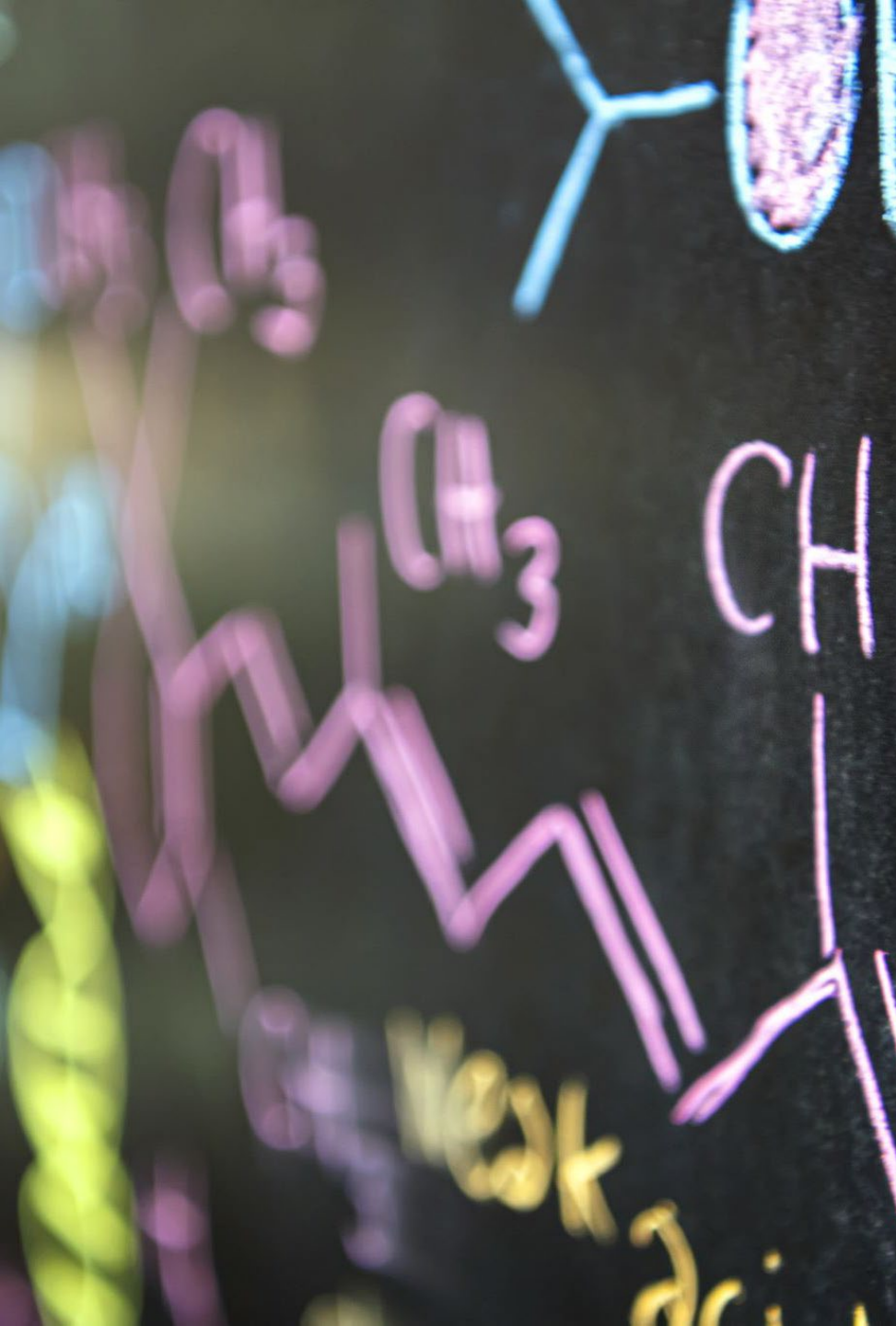
Combined Events

Types of combined events

- And events
- Or events

Prob of A and B

- If we have A and B as statistically independent.
 - $P(A \text{ and } B) = P(A)P(B)$
 - A = Ace, B = club
 - $P(A \text{ and } B) = 4/52 * 13/52 = 1/52$
 - Can be extended to more than two events, provided that all events are independent.
- If A and B aren't independent then
 - $P(A \text{ and } B) = P(A)P(B | A) = P(B)P(A | B)$
 - $4/52 * 1/4 = 1/52$

A chalkboard with colorful mathematical scribbles in the background. The scribbles include a blue 'Y' shape, a purple oval, a blue '3', a purple 'CH', a blue 'W', a purple 'Z', and a yellow '2'.

Prob of A or B

- If A and B are disjoint
 - $P(A \text{ or } B) = P(A) + P(B)$
 - A = Club, B = diamond
 - $13/52 + 13/52 = 26/52 = \frac{1}{2}$
- If not disjoint,
 - $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
 - A = club, B = king
 - $P(A \text{ or } B) = 13/52 + 4/52 - 1/52 = 16/52$

Wrapping Up...
